Module 6 Homework Report

Student name: PO-YING, HO

Branch name: clean-poying

Branch URL: https://github.com/jiamaotian/misinformation-analysis/tree/clean-poying

Remote: https://github.com/jiamaotian/misinformation-analysis.git

Submission commit ID: d7d7f6a2cd8fc81c08d7438ae1c9fc530b390f00

This report documents the completed Module 6 cleaning, validation, reproducibility, and submission evidence.

Work Attribution

The student reviewed and confirmed the expected outcomes for the five difficult manual cases before actual cleaned outputs were checked.

Codex performed the mechanical repo work: reading project files, querying cleaned Parquet/audit outputs, rerunning the full pipeline, comparing reproducibility outputs, creating the small submission/ folder, and drafting this report. Statements such as "queried", "checked", or "reran" refer to Codex-executed checks unless explicitly described as student review.

This report does not claim that the student personally ran DuckDB queries or the reproducibility run.

Data Source, Version, And Date Notes

Data source: public X/Twitter Community Notes download ZIPs.

Local raw path used by the completed run: /Users/bobo/dso576/project/raw/communitynotes/data.

Raw source files used: 17 ZIP files across five tables: batSignals, notes, ratings, noteStatusHistory, and userEnrollment.

Important date distinction:

• The run manifest records local raw file modification times from 2026-10-02T22:57:14.561223+00:00 to 2026-10-02T23:03:44.514467+00:00.

• These modification times are not necessarily download dates.

• Actual download date reported by the student: 2026-09-15.

• The exact upstream Community Notes dataset version was not independently confirmed.

• Full cleaning run time was 2026-10-03T05:16:25.665994+00:00.

Raw ZIPs, full cleaned Parquet files, and full row-level audit outputs remain local-only under Git-ignored data/.

Project Vision, Research Question, Unit, And Metrics

This project keeps both directions documented in objectives.md.

Coverage direction: A Community Notes Product Manager at X wants to identify which posts or note requests need targeted note-writing effort to reduce gaps in Community Notes coverage.

Response Time direction: A Community Notes Product Manager at X wants to identify which types of notes need additional rating effort to reduce time required to reach a clear status.

Analysis units retained by the cleaning output:

• Request/post level: batSignals, keyed by tweetId.

• Note level: notes and noteStatusHistory, keyed by noteId.

• Rating level: ratings, keyed by (noteId, raterParticipantId) for candidate duplicate checks.

• Contributor snapshot level: userEnrollment, keyed by participantId.

Candidate metrics from project documents:

• Coverage: eligible request coverage rate, uncovered request rate, coverage gap between groups, eligibility-to-first-note time.

• Response Time: helpful display rate, median time to first Helpful or first non-NMR status, long-wait unresolved rate, status reversal rate.

Confirmed direction and unresolved analysis choices:

• The project will keep both Coverage and Response Time as active directions.

• Feed choice is unresolved.

• Observation period is unresolved.

• Cutoff date is unresolved.

• Final metric denominators are unresolved.

• For Response Time, the endpoint is unresolved: first non-NMR, first Helpful, locked status, or another event.

Teacher Checklist

#	Item	Status	Evidence and notes
1	專題願景與問題	Done	objectives.md defines both Coverage and Response Time directions; this report preserves both.
2	每列定義	Done	Table grains are documented above and preserved in the cleaner. Evidence: plan.md, scripts/clean_communitynotes.py, validation row counts below.
3	保留來源	Done	Every cleaned row has _source_file, _source_member, _source_row; raw ZIP hashes and full source reconstruction were checked.
4	欄位與型態	Done	All 98 source columns have explicit rules; schema/type comparisons passed in reproducibility checks.
5	文字標準化	Done	Original text is retained; trimmed/list/parsed companions are added. Evidence: decision log and manual cases.
6	重複資料	Done	Candidate-key duplicate groups and exact duplicate rows were checked separately; both are zero for all five tables.
7	缺失資料	Done	Missing and sentinel handling is documented and checked. Before/after missing counts for affected fields are in Appendix A.
8	可疑值	Done	Suspicious values are flagged before treatment and retained when unresolved. Evidence: suspicious-value table and manual cases.
9	合併驗證	Done	Join keys, expected relationships, before/after row counts, and unmatched counts are listed below.
10	筆數與總數核對	Done	Source and cleaned row counts match for all tables; ratings summary reconciles to all ratings.
11	計算與分母	Unresolved	Some denominators are verified, such as valid helpfulnessLevel; final Coverage/Response Time denominators remain unresolved.
12	五筆紀錄核對	Done	Five student-reviewed expected outcomes and Codex-queried actual results are included below.
13	可重現性	Done	Fresh full run passed validation; old/new outputs had matching file lists and content fingerprints, with only metadata/order differences.
14	剩餘限制	Done	Remaining unresolved items are listed at the end.

Row Counts And Duplicate Checks

Different duplicate types are separated:

• Candidate-key duplicate groups: repeated logical keys such as noteId or (noteId, raterParticipantId).

• Exact duplicate rows: complete source-record copies.

• Analysis exclusions: exact duplicate copies removed from analysis views, if any.

Table	Candidate key	Raw rows	Cleaned rows	Candidate-key duplicate groups	Exact duplicate rows	Analysis exclusions
batSignals	tweetId	487,851	487,851	0	0	0

Table	Candidate key	Raw rows	Cleaned rows	Candidate-key duplicate groups	Exact duplicate rows	Analysis exclusions
notes	noteId	3,137,735	3,137,735	0	0	0
ratings	noteId, raterParticipantId	235,530,477	235,530,477	0	0	0
noteStatusHistory	noteId	3,343,818	3,343,818	0	0	0
userEnrollment	participantId	1,497,640	1,497,640	0	0	0

Evidence: data/module6_validation/*-keys.json, submission/evidence/validation_summary.md, submission/evidence/run_summary.json.

Field Types, Type Conversion Failures, And Missing Values

All source rows were retained. Type conversion changed storage type for many values, for example 0/1 to booleans and epoch integers to timestamp companions. These are type changes, not row drops.

Verified type-conversion result:

• Mapping mismatches against the recorded SQL contract: 0 across all checked source columns.

• Nonblank boolean/integer/timestamp parse-to-null failures in the full-column comparison: 0.

• Known sentinel-to-null typed values are not counted as parse failures. Examples: notes.tweetId=-1, ratings.ratedOnTweetId=-1, and ratings.suggestionId=-1.

• Suspicious or invalid values remain retained and flagged. Example: userEnrollment.timestampOfLastStateChange has two flag events on two different rows: invalid_timestamp = 1 (source row 687,733, raw value 0) and suspicious_timestamp_range = 1 (source row 1,224,917, raw value 1736527963).

Affected missing-value examples are in Appendix A. The most important ones:

Field	Raw missing or sentinel condition	Cleaned missing/NULL	Handling
notes.tweetId	8 rows with sentinel -1	8 NULL usable IDs	Note retained; unavailable for tweetId joins
notes.classification	2 raw blanks	2 NULL	Note retained; classification unknown
ratings.helpfulnessLevel	34,394 raw blanks	34,394 NULL	Legacy ratings retained; not in valid-level denominator
ratings.ratedOnTweetId	2,631,448 sentinel/unusable IDs	2,631,448 NULL	Rating retained; unavailable for this join key
ratings.suggestionId	235,514,406 sentinel/unusable IDs	235,514,406 NULL	Rating retained; original token recoverable
noteStatusHistory.timestampMillisOfFirstNonNMRStatus	2,767,657 raw blanks	2,767,657 NULL	Status row retained; unavailable for completed first-non-NMR timing
userEnrollment.timestampOfLastStateChange	0 raw blanks; 1 invalid timestamp flag and 1 suspicious range flag on two different rows	typed integer retained; UTC companion has 1 missing/unusable value in column comparison	Retained and flagged; not auto-repaired

Suspicious Values

Flag events can overlap on the same row, so event counts are not the same as distinct row counts.

Table	Flag events	Distinct flagged rows	Main rules
batSignals	9	5	malformed source URLs and nested source URLs
notes	4	4	missing classification and blank text
ratings	13	13	blank suggestion text
noteStatusHistory	2,175,237	2,174,994	unverified -1 stable-NMR semantics; resolved missing first history
userEnrollment	549	549	earn-out disagreement = 547; timestampOfLastStateChange:invalid_timestamp = 1; timestampOfLastStateChange:suspicious_timestamp_range = 1

Handling: suspicious values are retained with original source references and conservative treatment. No row is deleted solely because it is suspicious.

Join Checks

Join validation preserved row counts. Unmatched rows were flagged, not dropped. These checks are not final Coverage or Response Time denominators.

Join	Key	Expected relation	Rows before	Rows after	Matched	Unmatched incl. unavailable	Unavailable left key
notes -> requests	tweetId	many notes to request/post lookup	3,137,735	3,137,735	170,383	2,967,352	8
notes -> status	noteId	note to status lookup	3,137,735	3,137,735	2,897,146	240,589	0
status -> notes	noteId	status to note lookup	3,343,818	3,343,818	2,897,146	446,672	0
notes -> authors	noteAuthorParticipantId=participantId	many notes to enrollment lookup	3,137,735	3,137,735	3,137,735	0	0
ratings -> notes	noteId	many ratings to note lookup	235,530,477	235,530,477	212,762,691	22,767,786	0
ratings -> raters	raterParticipantId=participantId	many ratings to enrollment lookup	235,530,477	235,530,477	235,530,477	0	0
requests -> notes	tweetId with distinct note tweet IDs	request/post to at-least-one-note lookup	487,851	487,851	111,372	376,479	0

Additional checks:

• Matched note/status creation-time disagreements: 0.

• Matched note/status author disagreements: 0.

• Matched ratings earlier than note creation: 0.

Evidence: data/module6_validation/joins.json, submission/evidence/validation_summary.md.

Verified Calculations And Effective Denominators

These are verified cleaning/validation denominators, not final project denominators.

helpfulnessLevel distribution:

Value	Count
HELPFUL	144,595,094
SOMEWHAT_HELPFUL	6,092,969
NOT_HELPFUL	84,808,020
Missing legacy level	34,394

Effective denominator for valid helpfulnessLevel distribution: 235,496,083. This equals version-2 rating count. It is not a Coverage denominator and not a Response Time denominator.

Helpful share among valid helpfulnessLevel ratings: 144,595,094 / 235,496,083 = 61.4002%.

Other verified denominators:

• Total ratings: 235,530,477.

• Distinct note groups in ratings_by_note: 3,215,297.

• Sum of rating_count in ratings_by_note: 235,530,477.

• Status rows: 3,343,818.

• Status rows with nonmissing first non-NMR time: 576,161.

• Currently resolved status rows: 419,954.

• Currently resolved rows missing first non-NMR time: 1,357.

Feed state counts are verified but final feed choice is unresolved:

Feed field	Eligible	Ineligible	Unknown
noteRequestFeedEligibleAtMillis	28,597	459,254	0
apiSmallFeedEligibleAtMillis	28,964	458,887	0
apiLargeFeedEligibleAtMillis	168,583	319,268	0
apiXlFeedEligibleAtMillis	487,506	345	0

Cleaning Decision Log

Area	Rule	Reason	Check
Raw files	Read ZIPs directly; do not extract or edit them	Preserve source integrity	Manifest SHA-256 verified before and after run
Provenance	Add _source_file, _source_member, _source_row to every row	Make every cleaned value traceable	Full TSV roundtrip checks
IDs	Store large IDs as strings	Avoid precision loss	Manual cases 1-3
ID sentinels	Known -1 usable-key sentinels become null typed IDs; raw tokens remain recoverable	Avoid false joins on -1	Manual cases 2-3
Booleans	Convert 0/1 to false/true; invalid values are flagged	Typed binary analysis without guessing	Unit tests and validation summary
Categories	Trim structured category labels; preserve unknown nonblank labels	Avoid speculative remapping	Rules and transform reports
Timestamps	Preserve original integers and add nullable UTC companions	Keep raw values auditable	Manual cases 4-5
PCRH signed time	Use absolute milliseconds for UTC companion and derive pcrh_state from sign	Sign contains state information	Manual case 4
Scoring minute	Keep timestampMinuteOfFinalScoringOutput as integer metadata	It is not an epoch event timestamp	Manual case 4
Text	Preserve original text; add trimmed companions	Avoid changing content while exposing whitespace issues	Validation summary text checks
URLs/nested suggestions	Split source links, parse nested JSON, convert valid nested IDs to strings, flag malformed URLs	Preserve structure without unsupported repair	Manual case 1
Duplicates	Keep base rows; exclude only exact original-record copies from analysis views	Avoid dropping conflicting data	Duplicate reports show zero exclusions
Joins	Preserve left-row cardinality and write join flags	Nonmatches may reflect snapshot scope	Join validation reports

Five Record Checks

The student reviewed and confirmed expected outcomes before actual cleaned results were queried. Codex later queried actual cleaned rows and audit flags. The expected results below are summarized from the previously confirmed expectations and were not rewritten to match the actual output.

1. batSignals**, row 14,509**

Raw value(s): sourceLinks="yuujihttps://x.com/..."; nested source_link="Yuujihttps://x.com/..."; small/standard feed -1; large/XL 1784756989691

Expected result and reason: Keep request and raw values; split source links but do not repair malformed prefixes; -1 feed values become ineligible/null UTC; large/XL become eligible UTC; nested suggestion_id becomes string to avoid precision loss.

Actual result: sourceLinks_list retained malformed token; suggestions_parsed has string "2080047309124174174"; standard/small ineligible; large/XL eligible at 2026-07-22 21:49:49.691000+00:00; URL flags recorded.

Match?: Yes

2. notes**, row 114**

Raw value(s): tweetId="-1"; blank classification, believable, harmful, validationDifficulty; isCollaborativeNote="1"

Expected result and reason: Keep note; make tweetId unavailable for joins; keep classification and related fields missing; convert binary fields; preserve summary text. Reason: missing classification and unavailable tweet ID do not invalidate note-level record.

Actual result: tweetId=NULL; classification-related fields NULL; isMediaNote=false; isCollaborativeNote=true; summary preserved; classification:missing_classification flag.

Match?: Yes

3. ratings**, row 1**

Raw value(s): version 1; helpful="1"; notHelpful="0"; blank helpfulnessLevel; ratedOnTweetId="-1"; suggestionId="-1"

Expected result and reason: Keep legacy rating; preserve helpful/notHelpful; do not infer helpfulnessLevel; exclude from valid-level denominator. Reason: legacy binary format is separate from new level format.

Actual result: version=1; helpful=true; notHelpful=false; helpfulnessLevel=NULL; ratedOnTweetId=NULL; suggestionId=NULL.

Match?: Yes

4. noteStatusHistory**, row 595**

Raw value(s): timestampMillisAbovePcrhThreshold="-1788540132611"; timestampMinuteOfFinalScoringOutput="29845679"; current status NMR but core/expansion Helpful

Expected result and reason: Preserve signed PCRH value; derive UTC from absolute value and no_longer_above; keep scoring minute as integer, not UTC; do not overwrite current overall status from submodels.

Actual result: PCRH UTC 2026-09-04 16:42:12.611000+00:00; pcrh_state="no_longer_above"; scoring minute has no _utc; current status remains NEEDS_MORE_RATINGS; NMR stable-time -1 flagged unresolved.

Match?: Yes

5. userEnrollment**, row 1,224,917**

Raw value(s): timestampOfLastStateChange="1736527963"; timestampOfLastEarnOut="1712159915303"; modelingGroup="13.0"

Expected result and reason: Keep participant; flag suspicious timestamp range/unit; do not multiply by 1000 or overwrite from earn-out time; parse exact integral group as 13. Reason: unit issue is plausible but unconfirmed.

Actual result: State-change integer retained; UTC companion 1970-01-21 02:22:07.963000+00:00; earn-out UTC remains separate; modelingGroup=13; suspicious timestamp flag recorded.

Match?: Yes, with analytic use unresolved

Codex Suggestion Check

Codex suggested that timestampOfLastStateChange=1736527963 in Case 5 should be treated as suspicious and not automatically converted as seconds or overwritten with timestampOfLastEarnOut.

Student role: reviewed the expected treatment and confirmed it should remain conservative.

Codex role: queried the cleaned row and audit flag. The query found that the pipeline retained the raw integer, produced a 1970 UTC companion, kept timestampOfLastEarnOut separate, and recorded suspicious_timestamp_range.

Conclusion: the Codex suggestion was checked against actual cleaned output and audit evidence. It remains an unresolved data-quality issue, not a repaired value.

Reproducibility Summary

Reproduction Commands

Run from the repository root. Replace the raw-data path with the folder containing the same 17 ZIP files. Choose a new, Git-ignored output directory so existing verified outputs are not overwritten.

```
uv venv .venv --python 3.13
uv pip install --python .venv/bin/python -r requirements-cleaning.txt
.venv/bin/python -m unittest discover -s tests -v
.venv/bin/python scripts/clean_communitynotes.py \
  --raw-dir /path/to/communitynotes/data \
  --output-dir data/repro_check_new \
  --workers 1
```

The completed verification used Python 3.13.2, duckdb 1.5.6, pyarrow 25.0.1, and pytz 2026.4. The actual verified rerun used the local raw path listed above and output directory data/repro_cleaning_run; the command here uses a new directory to prevent overwriting it. Expected outputs include manifest.json, cleaned Parquet partitions, audit files, summaries, catalog.sql, and reports/05_validation.json. The completed rerun reported status=passed, and all 11 unit tests passed. The comparisons below are evidence from that completed run; these instructions have not triggered another run.

Codex executed a fresh full run to data/repro_cleaning_run without overwriting data/cleaning_run.

Results:

• New run validation: passed.

• File list: 86 old files and 86 new files; no missing or extra files.

• Parquet files compared: 35.

• Schema/type mismatches: none.

• Row count mismatches: none.

• Order-insensitive content fingerprint mismatches: none.

• Differences observed: run metadata, run-specific paths, JSON list ordering, and row order in four SQL-derived outputs.

Evidence copied into the submission folder:

• submission/evidence/reproducibility_report.md

• submission/evidence/reproducibility_comparison_summary.json

Sample Data

The Git-safe cleaned sample contains 5 total data rows, one per table, selected from the five manual difficult cases.

Files:

• submission/cleaned_sample/batSignals_manual_case_sample.csv

• submission/cleaned_sample/notes_manual_case_sample.csv

• submission/cleaned_sample/ratings_manual_case_sample.csv

• submission/cleaned_sample/noteStatusHistory_manual_case_sample.csv

• submission/cleaned_sample/userEnrollment_manual_case_sample.csv

Selection method and source rows are documented in submission/sample_manifest.json.

Appendix A: Missing Counts For Affected Fields

Fields with raw blanks, sentinel-to-null handling, UTC companion missingness, or validation-relevant NULL counts are listed here. Mapping mismatches were 0. This appendix is based on data/module6_validation/column_comparison.md; no data was rerun for this update.

Table	Field	Raw empty count	Cleaned NULL count	Nonblank parse-to-null failures	Notes
batSignals	noteRequestFeedEligibleAtMillis_utc	0 in source field	459,254	0	UTC companion null when feed is ineligible
batSignals	apiSmallFeedEligibleAtMillis_utc	0 in source field	458,887	0	UTC companion null when feed is ineligible
batSignals	apiLargeFeedEligibleAtMillis_utc	0 in source field	319,268	0	UTC companion null when feed is ineligible
batSignals	apiXlFeedEligibleAtMillis_utc	0 in source field	345	0	UTC companion null when feed is ineligible
batSignals	sourceLinks	372,919	0	N/A	Text preserved; empty text remains empty; list companion can be empty
notes	tweetId	0	8	N/A	-1 sentinel made unusable typed ID
notes	classification	2	2	N/A	Missing classification retained
notes	believable	3,113,054	3,113,054	N/A	Missing category retained
notes	harmful	3,113,054	3,113,054	N/A	Missing category retained
notes	validationDifficulty	3,113,054	3,113,054	N/A	Missing category retained
ratings	helpfulnessLevel	34,394	34,394	N/A	Legacy ratings retained; no imputation
ratings	ratedOnTweetId	0	2,631,448	N/A	-1 sentinel made unusable typed ID
ratings	suggestion	235,503,178	0	N/A	Text preserved; blank text remains blank
ratings	suggestionId	0	235,514,406	N/A	-1 sentinel made unusable typed ID
noteStatusHistory	timestampMillisOfFirstNonNMRStatus	2,767,657	2,767,657	0	Missing milestone retained
noteStatusHistory	firstNonNMRStatus	2,767,657	2,767,657	N/A	Missing milestone category retained
noteStatusHistory	timestampMillisOfLatestNonNMRStatus	2,767,657	2,767,657	0	Missing latest non-NMR time retained
noteStatusHistory	mostRecentNonNMRStatus	2,767,657	2,767,657	N/A	Missing latest non-NMR status retained
noteStatusHistory	timestampMillisOfStatusLock	240,179	240,179	0	Missing lock time retained
noteStatusHistory	lockedStatus	240,179	240,179	N/A	Missing lock status retained

Table	Field	Raw empty count	Cleaned NULL count	Nonblank parse-to-null failures	Notes
noteStatusHistory	timestampMillisOfRetroLock	3,342,306	3,342,306	0	Missing retro-lock time retained
noteStatusHistory	currentCoreStatus	706,547	706,547	N/A	Missing submodel status retained
noteStatusHistory	currentExpansionStatus	183,861	183,861	N/A	Missing submodel status retained
noteStatusHistory	currentGroupStatus	2,289,825	2,289,825	N/A	Missing group status retained
noteStatusHistory	currentModelingGroup	2,289,825	2,289,825	0	Missing group id retained
noteStatusHistory	timestampMillisOfMostRecentStatusChange_utc	0 in source field	1,093,544	0	UTC companion null for -1 sentinel/no recorded change
noteStatusHistory	timestampMillisOfNmrDueToMinStableCrhTime	1,170,025	1,170,025	0	Blank retained as missing; -1 semantics flagged separately
noteStatusHistory	timestampMillisOfNmrDueToMinStableCrhTime_utc	1,170,025 source blanks plus -1/unavailable values	3,343,787	0	UTC companion null for blanks and sentinel/unavailable values
noteStatusHistory	currentMultiGroupStatus	3,280,453	3,280,453	N/A	Missing multigroup status retained
noteStatusHistory	currentModelingMultiGroup	3,280,453	3,280,453	0	Missing multigroup id retained
noteStatusHistory	timestampMillisOfFirstNmrDueToMinStableCrhTime	3,112,207	3,112,207	0	Blank retained as missing; -1 semantics flagged separately
noteStatusHistory	timestampMillisOfFirstNmrDueToMinStableCrhTime_utc	3,112,207 source blanks plus -1/unavailable values	3,112,325	0	UTC companion null for blanks and sentinel/unavailable values
noteStatusHistory	timestampMillisAbovePcrhThreshold	3,317,870	3,317,870	0	PCRH absent vs signed values handled separately
noteStatusHistory	timestampMillisAbovePcrhThreshold_utc	3,317,870 source blanks plus unavailable/invalid PCRH values	3,324,230	0	UTC companion null when no recorded event or invalid/unavailable
userEnrollment	timestampOfLastStateChange	0	0	0	Raw integer retained
userEnrollment	timestampOfLastStateChange_utc	0 source blanks	1	0	One invalid value (row 687,733; raw 0) has NULL UTC. A different row (1,224,917) retains a 1970 UTC value and has a suspicious-range flag.
userEnrollment	timestampOfLastEarnOut	0	0	0	Raw integer retained, including sentinel
userEnrollment	timestampOfLastEarnOut_utc	0 source blanks	1,407,922	0	UTC companion null for documented never-earned-out sentinel

Table	Field	Raw empty count	Cleaned NULL count	Nonblank parse-to-null failures	Notes
userEnrollment	never_earned_out_consistent	derived field	547	N/A	Null where earn-out sentinel and count disagree

Remaining Limitations And Missing Information

• Submission commit ID: d7d7f6a2cd8fc81c08d7438ae1c9fc530b390f00

• Exact upstream dataset version: TBD / not confirmed.

• Actual download date: 2026-09-15, as reported by the student. Local raw file modification times are recorded separately and are not claimed as download dates.

• Coverage and Response Time are both retained as confirmed project directions.

• Feed choice, observation period, cutoff date, and final metric denominators remain Unresolved.

• Response Time endpoint remains Unresolved: first non-NMR, first Helpful, locked status, or another event.

• Some sentinel semantics remain unresolved, especially NMR stable-time -1 meanings.

• Suspicious timestamp unit issues are retained and flagged but not resolved.

• Cross-table nonmatches are preserved and flagged; their real-world causes are not proven by the available snapshots.